
GLOBAL COMMODITY INSIGHTS LTD

Commodity Market Intelligence Platform

Data Understanding Report

Project	Commodity Market Intelligence Platform
Team	Ajay Khadke Akshay Londhe
Platform	Snowflake Cloud Analytics (COMMODITY_WH)
Datasets	gold_stocks_price.csv oil_price.csv
Coverage	December 2011 to December 2018
Date	June 2026

1. Executive Summary

This report documents the data understanding phase of the Commodity Market Intelligence Platform. The Data Engineering team has sourced, profiled, and assessed two market datasets covering Gold and Oil prices across a seven-year period from December 2011 to December 2018. The findings below provide the business context, structural profile, and quality baseline required before data pipeline construction begins.

Together the two files hold 3,562 trading records across 87 columns. Every critical field passed quality checks with zero nulls and zero duplicates, so the data is ready for time-series analysis and dashboard reporting.

2. Business Context

2.1 Organisation Background

Global Commodity Insights Ltd is a financial analytics company delivering market intelligence to investment firms, commodity traders, business analysts, and executive decision-makers. The organisation currently faces significant operational inefficiencies caused by data stored in spreadsheets, manual reporting processes, and no centralised platform for analysis.

2.2 Business Problem

The Executive Leadership Team has identified the following challenges that this data platform aims to resolve:

- Data received from multiple disconnected sources with no single source of truth
- Market data stored in CSV and spreadsheet formats with no version control
- Reporting is largely manual, consuming analyst time that should be spent on insights
- Data quality issues are difficult to identify and track systematically
- Business users cannot easily answer time-sensitive questions about market trends
- No central platform exists for storing, querying, or visualising commodity data

2.3 Strategic Questions This Data Enables

The datasets directly support the following business questions:

- How have Gold and Oil prices changed over the seven-year period?
- Are there significant market events visible in trading volumes or price volatility?
- How do Gold prices correlate with equity market indices (S&P 500, Dow Jones)?
- Can we identify periods of market stress and quantify their impact?
- What trends should be highlighted in executive dashboards?

3. Dataset Overview

3.1 Dataset 1: Gold and Financial Markets

File: gold_stocks_price.csv | Source: AWS DataSpark Repository

Attribute	Detail
Row Count	1,718 trading days
Column Count	81 columns
Date Range	December 2011 to December 2018
Primary Commodity	Gold (GLD ETF proxy: Open, High, Low, Close, Volume)
Market Indices	S&P 500 (SP_), Dow Jones (DJ_)
ETF Data	Gold Miners ETF (GDX_), Oil ETF (USO_)
Currency / FX	US Dollar Index (USDI_), Euro (EU_)
Other Commodities	Oil Futures (OF_), Silver Futures (SF_), Platinum (PLT_), Palladium (PLD_), Rhodium (RHO_)
Metadata Columns	Trend indicators (_Trend) for several series
Date Format	DD/MM/YYYY
Encoding	UTF-8, comma-separated

3.2 Dataset 2: Oil Market Prices

File: oil_price.csv | Source: AWS DataSpark Repository

Attribute	Detail
Row Count	1,844 trading days
Column Count	6 columns
Date Range	December 2011 to December 2018
Columns	Date, Price (Close), Open, High, Low, Vol. (Volume)
Price Units	USD per barrel (WTI Crude Oil)
Volume Format	Abbreviated string (e.g. "275.86K"), needs parsing
Date Format	YYYY-MM-DD
Encoding	UTF-8, comma-separated

4. Column Inventory

4.1 Gold Dataset: Column Groups

The 81 columns in gold_stocks_price.csv are organised into the following logical groups:

Group	Columns	Description
Gold (Primary)	Date, Open, High, Low, Close, Adj Close, Volume	Core gold price OHLCV data, the central analytical series
S&P 500	SP_open, SP_high, SP_low, SP_close, SP_Ajclose, SP_volume	Daily S&P 500 index values for equity market context
Dow Jones	DJ_open ... DJ_volume (6 cols)	Dow Jones Industrial Average prices and volume
Euro Gold (EG)	EG_open ... EG_volume (5 cols)	Gold priced in Euro for FX cross-commodity context
Euro FX (EU)	EU_Price, EU_open, EU_high, EU_low, EU_Trend	Euro/USD exchange rate and directional trend
Oil Futures (OF)	OF_Price, OF_Open, OF_High, OF_Low, OF_Volume, OF_Trend	Crude oil futures prices for cross-commodity comparison
Silver (SF)	SF_Price, SF_Open, SF_High, SF_Low, SF_Volume, SF_Trend	Silver futures for precious metals peer comparison
Other Metals	PLT_, PLD_, RHO_PRICE (7 cols)	Platinum, Palladium and Rhodium rare metals data
US Dollar Index	USDI_Price, USDI_Open ... USDI_Trend (6 cols)	USD strength index, a key macro driver for gold prices
Gold ETF (GDX)	GDX_Open, GDX_High, GDX_Low, GDX_Close, GDX_Adj Close, GDX_Volume	Gold Miners ETF, a tradeable proxy for gold exposure
Oil ETF (USO)	USO_Open, USO_High, USO_Low, USO_Close, USO_Adj Close, USO_Volume	Oil ETF, a tradeable proxy for oil market exposure

4.2 Oil Dataset: Column Summary

Column	Type (Raw)	Type (Staged)	Notes
Date	VARCHAR	DATE	YYYY-MM-DD format, parses directly via TRY_TO_DATE
Price	VARCHAR	DOUBLE	WTI crude close price in USD per barrel
Open	VARCHAR	DOUBLE	Opening price for the trading session

High	VARCHAR	DOUBLE	Intraday high, used in OHLC validation
Low	VARCHAR	DOUBLE	Intraday low, used in OHLC validation
Vol.	VARCHAR	NUMBER	Abbreviated string (e.g. "275.86K"), needs REPLACE and a cast

5. Statistical Profile

5.1 Gold Price Summary

Metric	Gold Close (USD)		
Statistic	Value	Year	Note
All-time High	\$173.61	4 Oct 2012	Highest close in the dataset
All-time Low	\$100.50	17 Dec 2015	Lowest close in the dataset
7-Year Average	\$127.32	2011-2018	Mean across all 1,718 trading days
Std Deviation	\$17.54	Full period	Moderate volatility across the seven years
2012 Annual Avg	\$161.87	2012	Peak year, highest annual average in the data
2015 Annual Avg	\$111.11	2015	Trough year, lowest annual average in the data

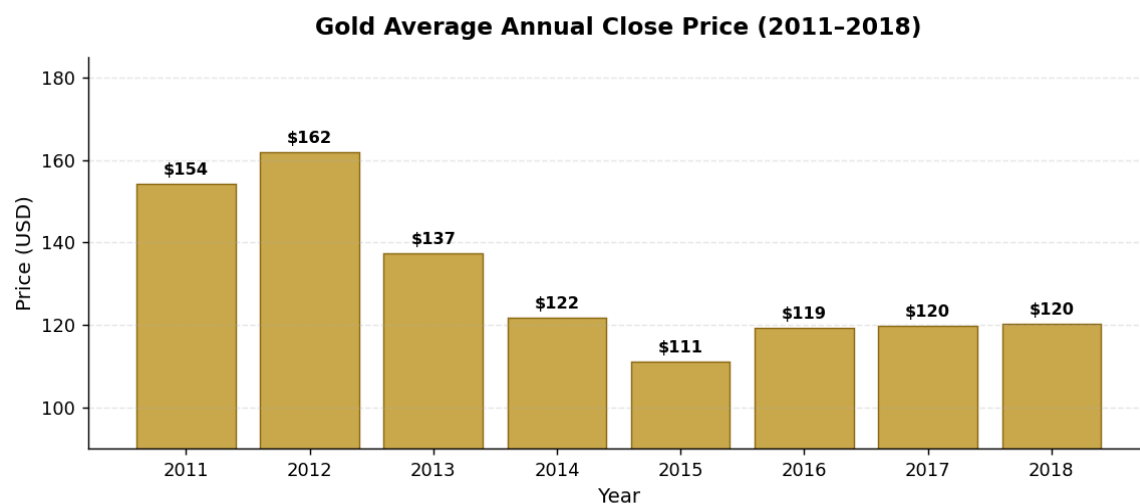


Figure 1: Gold Average Annual Close Price, 2011 to 2018

5.2 Oil Price Summary

Statistic	Value	Year	Note
All-time High	\$110.53	6 Sep 2013	Highest price before the supply-driven crash
All-time Low	\$26.21	11 Feb 2016	Historic trough during the oversupply period
7-Year Average	\$70.44	2011-2018	Mean across all 1,844 trading days
2013 Annual Avg	\$98.05	2013	Highest annual average in dataset
2016 Annual Avg	\$43.41	2016	Lowest full-year average in dataset
Crash Drawdown	-76.3%	2013-2016	From \$110.53 (Sep 2013) to \$26.21 (Feb 2016)

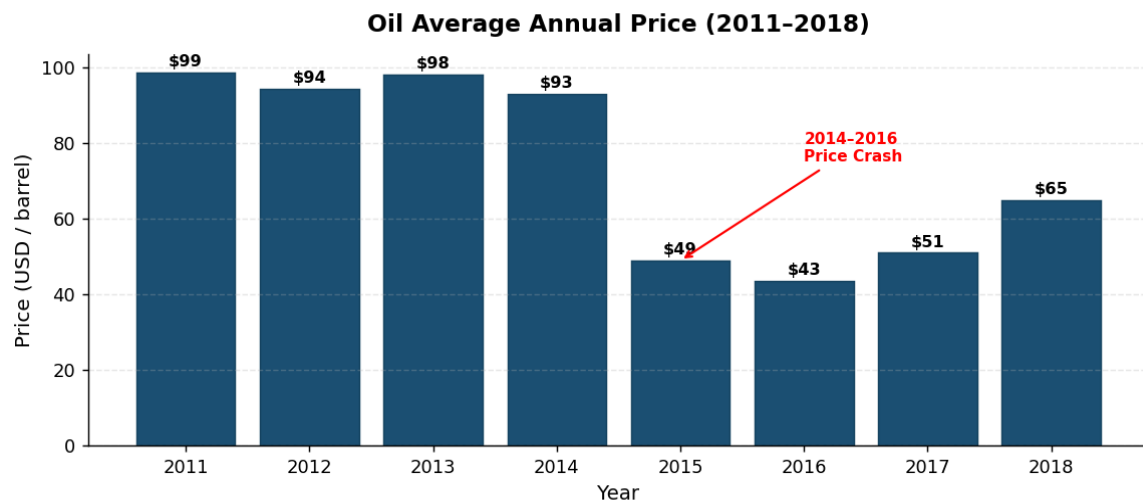


Figure 2: Oil Average Annual Price, 2011 to 2018

5.3 Combined Trend Overview

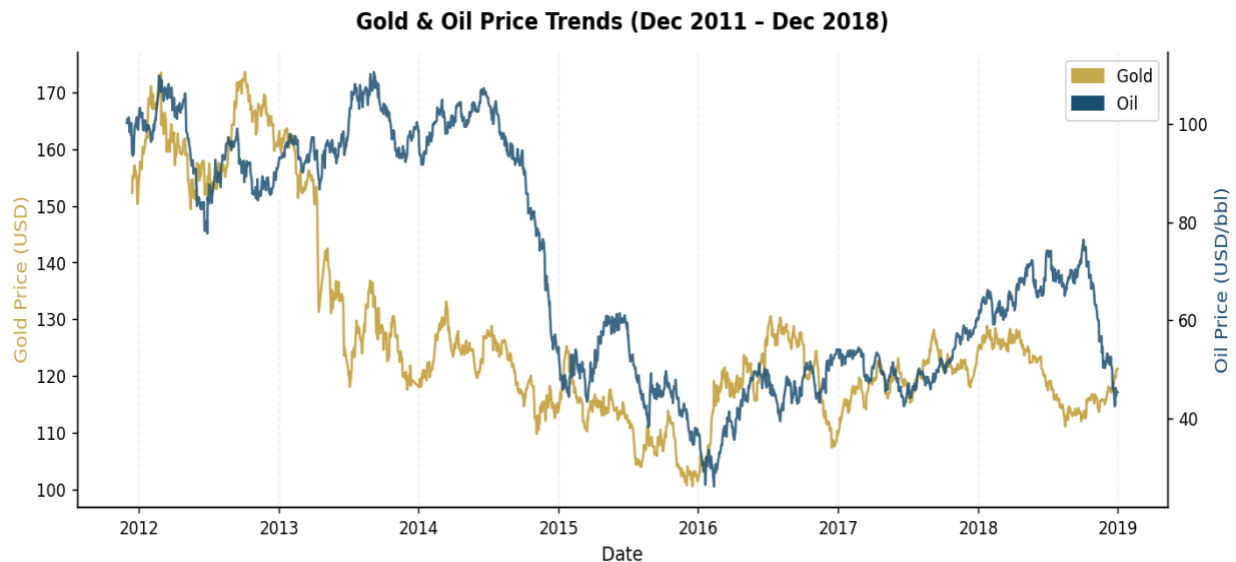


Figure 3: Gold and Oil Daily Price Trends (Dec 2011 to Dec 2018)

The trend chart shows gold and oil parting ways from mid-2014. Oil fell hard between 2014 and 2016 while gold held a fairly narrow range, which suggests the two markets were responding to different drivers over that stretch.

6. Initial Observations

6.1 Who Uses This Data

- Portfolio managers, who track gold as a defensive allocation alongside equities
- Commodity traders, who watch oil momentum and unusual volume for trading signals
- Risk analysts, who measure drawdowns and correlations for portfolio exposure
- Executive leadership, who need summarised trend views for quarterly reporting
- Data engineers, who need typed, deduplicated data to feed downstream analytics

6.2 Business Decisions Supported

- When to increase or reduce gold exposure relative to equities
- Whether the oil market crash represents a structural or cyclical shift
- How USD strength correlates with gold prices over the business cycle
- Identifying unusual trading volume days that may signal market events

6.3 Additional Data That Would Improve Analysis

- Macroeconomic series such as CPI, interest rates and central bank announcements
- A geopolitical event calendar to map price spikes to real-world catalysts
- News or sentiment data to support future modelling work
- Forward curves / futures term structure for oil
- More recent data beyond 2018 to validate whether historical patterns persist

Worth noting: the gold file is far richer (81 columns against 6) and supports much wider analysis, while the oil file is lean but clean and well suited to trend and crash work. Neither file contains duplicate dates or nulls in the price fields.

7. Snowflake Architecture Notes

This data has been loaded into a Snowflake cloud environment using a layered schema architecture: RAW > STAGING > ANALYTICS > AUDIT. The key design decisions are documented below.

Schema	Purpose	Key Tables
RAW	Unmodified ingested data; VARCHAR types with a full audit trail	RAW_GOLD_MARKET_DATA, RAW_OIL_MARKET_DATA
STAGING	Type-cast, deduplicated, OHLC-validated records	STG_GOLD_MARKET, STG_OIL_MARKET
ANALYTICS	Business-ready facts with moving averages, volatility and date attributes	FACT_GOLD_MARKET, FACT_OIL_MARKET, DIM_DATE and views
AUDIT	Persistent data quality log with severity and percentage affected	DQ_LOG

7.1 Automated Pipeline and Dashboard

The platform does not depend on anyone running scripts by hand. A Snowflake Stream watches the raw gold table for newly inserted rows, and a scheduled Task chain reacts automatically: it cleans and loads the new rows into staging, rebuilds the analytics fact table, and re-runs the data quality checks into the audit log. Because views always read live data, the executive dashboard reflects new data the moment the chain completes.

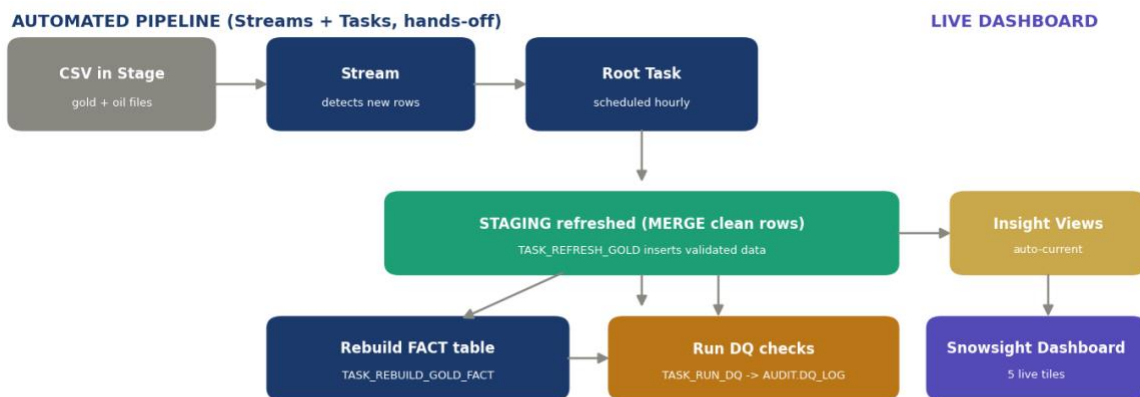


Figure 4: Automated Stream and Task pipeline feeding a live Snowsight dashboard

This design satisfies the project requirement for a repeatable ingestion process. The Task only consumes warehouse credits on cycles where the Stream actually contains new data, so an idle pipeline costs almost nothing. The resulting Snowsight dashboard surfaces five tiles: annual volatility, the gold and oil trend, regime correlation, data quality status, and the annual gold summary.

8. From Understanding to Hypotheses

Understanding the data is the starting point, not the destination. The structure and quality established here set up the analytical work that follows in Deliverable 4, where the team moves from describing the data to interrogating it. Rather than simply reporting prices and volumes, that report frames the questions a commodity desk would actually ask and tests each one against the seven years of evidence.

The hypotheses the data is well positioned to answer include:

- Is oil structurally more volatile than gold, and by how much?
- Does a strong US dollar genuinely push gold prices down?
- Does gold behave as a safe haven when equity markets sell off?
- Do gold and oil move together, or does holding both reduce risk?
- Can trading volume or volatility clustering act as an early warning signal?

Because the augmentation layer already computes daily returns, moving averages, rolling volatility and the market indices needed for these tests, the platform is ready to answer them directly in SQL. The verdicts, several of which challenge conventional assumptions, are presented in the hypothesis-driven insights report.

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